Case Study 1
BAN 673 – TIME SERIES ANALYTICS

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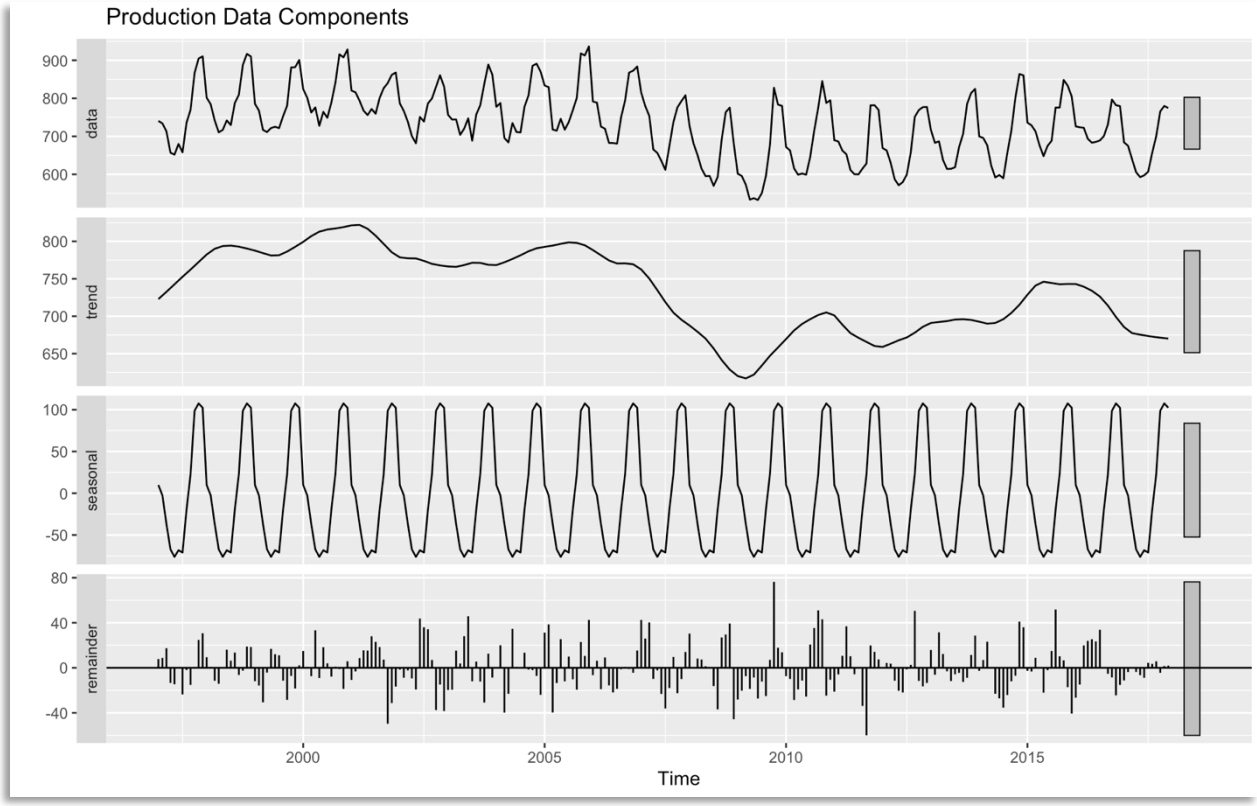
Case Study #1: Production Forecasting

Question 1b:

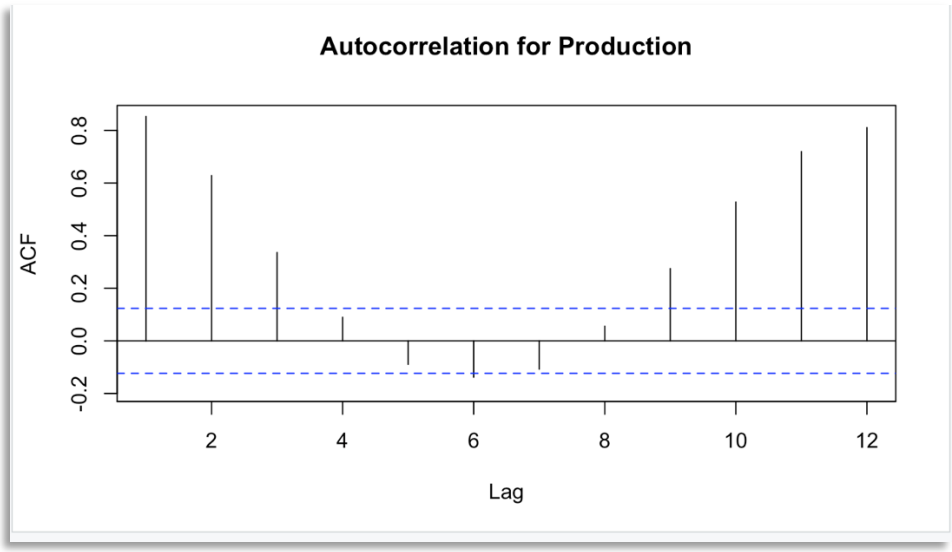


1. **Data:**
   The first graph shows the original time series data, we can see that there are clear repetitive ups and downs within each year this can mean that there is some seasonality in the data. Overall, production fluctuates between 500 (the lowest point in the graph) and 900 million tons approximately.
2. **Trend:**
   The trend component reveals long-term movements in the data. From 1997 to around 2004, production increased steadily. Then, from 2005 to 2010, there's a decline, followed by some fluctuations with a slight upward recovery until around 2015, after which it declines again.
3. **Seasonality:**
   The third graph "seasonal" is where we have extracted the seasonal component of the data and it is evident that there is seasonality in the data because of how we observe a consistent pattern. At the beginning of every year, the production starts from midway and gradually goes down and then towards the middle of the year the production gradually rises to reach peak and then towards the end the production drops. This pattern is seen across all the years and hence it is evident that there is consistent seasonality.
   In context of our case study this can indicate that the peak production is observed in summer of every year that is towards the mid-year.

And the fact that the magnitude of seasonality is consistent throughout the data makes it "Additive Seasonality".

4. **Remainder (Residual):**
   The remainder is the combination of Level and Randomness and it is random for different time periods. We observe irregular fluctuations not explained by trend or seasonality. Most residuals fall between -40 and +40, with occasional outliers. These could be attributed to unexpected shocks like weather anomalies, policy changes, or economic events.

Question 1c:



	Lag	ACF
1	0	1.000
2	1	0.853
3	2	0.628
4	3	0.336
5	4	0.090
6	5	-0.089
7	6	-0.139
8	7	-0.108
9	8	0.056
10	9	0.275
11	10	0.528
12	11	0.720
13	12	0.811

Autocorrelation Analysis:

The autocorrelation plot and ACF table reveal important characteristics of the production data.

Trend:
The high autocorrelation at lag 1 (ACF = 0.853) indicates a strong positive correlation between consecutive months, which suggests the presence of a trend in the data. This means that values in the series tend to follow a predictable direction over time. As we move to higher lags, the autocorrelation coefficients decline, even turning negative around lag 6–7, before rising again. This cyclical behavior often reflects underlying seasonal or periodic patterns.

Seasonality:
A notable peak appears again at lag 12 (ACF = 0.811), which indicates a strong yearly correlation. This suggests that values in the same month across different years (e.g., January 2005 vs. January 2006) are highly correlated, a clear sign of **annual seasonality**. This aligns with the seasonal patterns observed in our graph in 1b, where production tends to peak around mid-year and dip at the start and end of the year.

Question 2b:

```
#QUESTION 2A

nValid <- 48
nTrain <- length(prod.ts) - nValid
train.ts <- window(prod.ts, start = c(1997, 1), end = c(1997, nTrain))
valid.ts <- window(prod.ts, start = c(1997, nTrain + 1),
                   end = c(1997, nTrain + nValid))

#QUESTION 2B

ma.trailing_4 <- rollmean(train.ts, k = 4, align = "right")
ma.trailing_4
ma.trailing_7 <- rollmean(train.ts, k = 7, align = "right")
ma.trailing_7
ma.trailing_10 <- rollmean(train.ts, k = 10, align = "right")
ma.trailing_10
```

Question 2c:

Trailing MA forecast for window width of 4 (Just the point forecasts without the confidence intervals)

```
> ma.trail_4.pred$mean
          Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct
2014 780.1289 755.9250 719.5197 677.5015 656.8639 639.9160 634.6757 646.2173 671.2697 712.0089
2015 784.0292 759.0454 722.0160 679.4986 658.4616 641.1941 635.6982 647.0353 671.9241 712.5324
2016 784.2973 759.2598 722.1875 679.6359 658.5714 641.2820 635.7685 647.0915 671.9691 712.5684
2017 784.3157 759.2746 722.1993 679.6453 658.5789 641.2880 635.7733 647.0954 671.9722 712.5709
          Nov      Dec
2014 757.0794 787.5294
2015 757.4983 787.8645
2016 757.5271 787.8875
2017 757.5290 787.8891
>
```

Question 2d:

Accuracy measures of the 3 trailing MA forecasts

```
> round(accuracy(ma.trail_4.pred$mean, valid.ts), 3)
            ME   RMSE   MAE   MPE  MAPE  ACF1 Theil's U
Test set 5.367 60.249 50.61 0.172 7.091 0.778     1.388
> round(accuracy(ma.trail_7.pred$mean, valid.ts), 3)
             ME   RMSE    MAE   MPE  MAPE  ACF1 Theil's U
Test set 14.185 81.486 67.582 0.972 9.407 0.839     1.903
> round(accuracy(ma.trail_10.pred$mean, valid.ts), 3)
          ME   RMSE    MAE    MPE  MAPE  ACF1 Theil's U
Test set 7.1 81.471 66.024 -0.161 9.293 0.827     1.938
>
```

- **RMSE (Root Mean Squared Error)** measures the average magnitude of forecast error. The lower the RMSE, the more accurate the model. Among the three models, the model with window 4 has the lowest RMSE (60.249), indicating it is the most accurate in predicting monthly food production. In context of our case study, it would mean that on average, the model's monthly food production forecast deviates by around 60.25 million tons, which is a relatively small error given national scale production.

- **MAPE (Mean Absolute Percentage Error)** represents the average percentage error between actual and forecasted values. Model with window width 4 again has the lowest MAPE (7.091), suggesting it has the smallest relative error. The model's forecasts are, on

average**,** within 7.09% of the actual monthly production values indicating reliable predictive accuracy in production planning.

Conclusion:

Based on both RMSE and MAPE, the 4-month trailing moving average model outperforms the 7- and 10-month models. Therefore, it is the best trailing MA forecast for this dataset during the validation period.

Question 3a:

```
Call:
tslm(formula = train.ts ~ trend + season)

Residuals:
     Min       1Q   Median       3Q      Max 
-135.084  -24.042    1.197   34.498   93.152 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 826.61611   12.21683  67.662  < 2e-16 ***
trend        -0.75284    0.05416 -13.900  < 2e-16 ***
season2     -13.34475   15.59824  -0.856  0.39333    
season3     -50.76450   15.59852  -3.254  0.00134 ** 
season4     -78.56567   15.59899  -5.037 1.09e-06 ***
season5     -84.24477   15.59965  -5.400 1.96e-07 ***
season6     -76.28076   15.60050  -4.890 2.14e-06 ***
season7     -80.27163   15.60153  -5.145 6.60e-07 ***
season8     -28.92344   15.60275  -1.854  0.06532 .  
season9      15.99105   15.60416   1.025  0.30676    
season10     93.20183   15.60576   5.972 1.12e-08 ***
season11    101.70343   15.60755   6.516 6.24e-10 ***
season12     98.03791   15.60952   6.281 2.22e-09 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 45.48 on 191 degrees of freedom
Multiple R-squared:  0.7702,	Adjusted R-squared:  0.7558 
F-statistic: 53.35 on 12 and 191 DF,  p-value: < 2.2e-16

>
```

Model Equation:

Forecast (Y) = Intercept + Trend + Season2d_2+ Season3d_3+ ……+ Season12d_{12}

Production = 826.61 – 0.75 – 13.34d_2+ ……+ 98d_{12}

Here, d_2,d_3 .....d_{12} are dummy variables which are 1 if the forecast is for that particular month or 0 if it is not (e.g. d_2 = 1 if it is February, if not it is 0)

Terms:

- **Intercept (826.62):** The baseline production when all predictors are 0 (i.e., at the start of the series in January). This is the reason we have 11 dummy variables for 12months because when all of them are 0, that would mean it is for January.
- **Trend (-0.75):** A negative coefficient indicating a slight overall downward trend in production over time.
- **Seasonal coefficients (season2 to season12):** Capture monthly deviations from January Several months including season4, season5, season6, season7, season10, season11, and season12 show significant effects ($p < 0.05$), confirming the presence of strong monthly seasonality.

The negative trend suggests a mild decline in production over the 17-year training period. Seasonal coefficients reveal higher production levels in late summer and early fall (months 10–12), and lower production in spring and early summer (months 4–7), consistent with agricultural cycles.

Forecast:

```
> trend.seas.pred <- forecast(trend.seas, h = nValid, level = 0)
> trend.seas.pred$mean
          Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct
2014 672.2846 658.1870 620.0144 591.4604 585.0285 592.2396 587.4959 638.0913 682.2529 758.7109
2015 663.2505 649.1529 610.9804 582.4264 575.9944 583.2056 578.4619 629.0572 673.2189 749.6768
2016 654.2165 640.1189 601.9463 573.3923 566.9604 574.1715 569.4278 620.0232 664.1848 740.6428
2017 645.1824 631.0849 592.9123 564.3583 557.9263 565.1375 560.3938 610.9891 655.1508 731.6087
          Nov      Dec
2014 766.4596 762.0413
2015 757.4256 753.0072
2016 748.3915 743.9732
2017 739.3575 734.9391
>
```

Question 3c

```
+                        "MA.Residuals.Fst", "Combined.Fst")
> valid.df
   Validationdata Regression.Fst MA.Residuals.Fst Combined.Fst
1         699.898        672.285           35.954      708.239
2         695.778        658.187           39.715      697.902
3         676.501        620.014           42.724      662.739
4         622.509        591.460           45.131      636.592
5         591.739        585.028           47.057      632.085
6         598.157        592.240           48.597      640.837
7         589.649        587.496           49.830      637.326
8         656.481        638.091           50.816      688.907
9         715.300        682.253           51.604      733.857
10        801.637        758.711           52.235      810.946
11        863.854        766.460           52.740      819.200
12        860.447        762.041           53.144      815.185
13        735.949        663.251           53.467      716.718
14        728.953        649.153           53.725      702.878
15        713.015        610.980           53.932      664.913
16        676.449        582.426           54.098      636.524
17        647.782        575.994           54.230      630.224
18        674.915        583.206           54.336      637.541
19        688.269        578.462           54.420      632.882
20        775.777        629.057           54.488      683.545
21        775.733        673.219           54.542      727.761
22        848.340        749.677           54.586      804.263
23        833.438        757.426           54.620      812.046
24        804.789        753.007           54.648      807.655
25        726.254        654.216           54.670      708.887
26        723.755        640.119           54.688      694.807
27        722.490        601.946           54.702      656.649
28        693.549        573.392           54.714      628.106
29        683.266        566.960           54.723      621.683
30        685.427        574.172           54.730      628.902
31        689.087        569.428           54.736      624.164
32        700.777        620.023           54.741      674.764
33        731.709        664.185           54.744      718.929
34        796.843        740.643           54.747      795.390
35        782.070        748.392           54.750      803.141
36        779.462        743.973           54.752      798.725
37        684.211        645.182           54.753      699.936
38        675.143        631.085           54.754      685.839
39        639.920        592.912           54.755      647.668
40        605.572        564.358           54.756      619.114
41        592.620        557.926           54.757      612.683
42        597.337        565.138           54.757      619.895
43        606.970        560.394           54.758      615.151
44        656.763        610.989           54.758      665.747
45        700.549        655.151           54.758      709.909
46        765.700        731.609           54.758      786.367
47        779.865        739.358           54.759      794.116
48        774.537        734.939           54.759      789.698
>
```

Accuracy Measures:

```
> round(accuracy(trend.seas.pred$mean, valid.ts), 3)
             ME   RMSE    MAE   MPE  MAPE  ACF1 Theil's U
Test set 63.805 72.734 63.805 8.873 8.873 0.838     1.704
> round(accuracy(fst.2level, valid.ts), 3)
             ME   RMSE    MAE   MPE  MAPE  ACF1 Theil's U
Test set 11.004 35.322 28.741 1.368 4.078 0.833     0.853
>
```

• **RMSE:** The lower the RMSE, the more accurate the model. Between the two models, the two-level model has the lowest RMSE **(**35.322**)** compared to 72.734 from the regression model with trend and seasonality alone.

• **MAPE:** The two-level model again performs better with a MAPE of 4.078%**,** compared to 8.873% for the single regression model.

Conclusion:
Based on both RMSE and MAPE, the two-level forecasting model (regression with trend and seasonality + trailing MA for residuals) outperforms the simple regression model. Therefore, it is the best forecasting model for the validation period in this case study.

Question 3d:

```
> names(future12.df) <- c("Regression.Fst", "MA.Residuals.Fst", "Combined.Fst")
> future12.df
   Regression.Fst MA.Residuals.Fst Combined.Fst
1         677.215           -0.217      676.998
2         664.723           -0.685      664.037
3         630.406           -1.060      629.347
4         599.965           -1.359      598.606
5         590.822           -1.599      589.223
6         598.585           -1.791      596.794
7         595.608           -1.944      593.664
8         646.844           -2.067      644.777
9         688.950           -2.165      686.785
10        764.618           -2.244      762.374
11        773.115           -2.307      770.808
12        767.633           -2.357      765.277
>
```

Question 3e:

```
> round(accuracy((snaive(prod.ts))$fitted, prod.ts), 3)
            ME   RMSE    MAE    MPE  MAPE  ACF1 Theil's U
Test set -3.93 48.394 38.897 -0.786 5.519 0.763     1.056
>
> round(accuracy(tot.trend.seas.pred$fitted, prod.ts), 3)
         ME   RMSE    MAE    MPE  MAPE  ACF1 Theil's U
Test set  0 46.194 36.103 -0.443 5.125 0.856     1.046
>
> round(accuracy(tot.trend.seas.pred$fitted+tot.ma.trail.res, prod.ts), 3)
            ME   RMSE    MAE    MPE  MAPE  ACF1 Theil's U
Test set 0.332 19.461 15.032 -0.023 2.054 0.369     0.411
>
```

• **RMSE:** Among the three models, the two-level model has the lowest RMSE **(**19.461**)** compared to 48.394 for the seasonal naïve forecast and 46.194 for the regression model.

In context of our case study, this means the two-level model provides monthly food production forecasts that deviate by only about 19.46 million tons on average making it a highly precise tool for national-level production forecasting.

• **MAPE:** The two-level model once again outperforms with a MAPE of just 2.054%, while the seasonal naïve and regression models have MAPE values of 5.519% and 5.125%**,** respectively.

This implies that the two-level model's predictions are, on average, within 2.05% of actual monthly production, which offers a strong degree of reliability for planning and decision-making in the food production industry.

Conclusion:
Based on the lower RMSE and MAPE, the two-level model (regression with trend and seasonality + trailing MA for residuals) provides the most accurate forecasts when using the full historical dataset. Therefore, it is the best forecasting model for predicting U.S. food production for the 12 months of 2018 in this case study.

Question 4a:

```
> hw.ZZZ <- ets(train.ts, model = "ZZZ")
> hw.ZZZ
ETS(A,N,A)

Call:
ets(y = train.ts, model = "ZZZ")

  Smoothing parameters:
    alpha = 0.6533
    gamma = 1e-04

  Initial states:
    l = 773.2707
    s = 104.6714 107.1962 100.0903 23.7739 -20.8932 -71.5641
           -67.8861 -74.2349 -67.8762 -40.0424 -2.6853 9.4505

  sigma:  25.5169

     AIC     AICc      BIC
2422.043 2424.597 2471.815
>
```

Holt-Winter's Model for the Training Partition

We use the Holt-Winter's Model to capture trend and seasonality in the training dataset. We used this with automated component selection (model = "ZZZ"). The model chosen by the algorithm was:

ETS (A, N, A)

- **Additive Error (A)**: The forecast errors are added to the predicted values.
- **No Trend (N)**: The model does not include a trend component.
- **Additive Seasonality (A)**: Seasonal effects are added.

Model Parameters:

- **Alpha (level smoothing)** = 0.6533: Moderate weighting of recent observations for level estimation.
- **Gamma (seasonal smoothing)** = 0.0001: Very low smoothing of seasonal components, indicating stable seasonality over time.
- **Sigma** = 25.52: Represents the standard deviation of the residuals (errors), showing the model's average forecasting uncertainty.

In the context of U.S. food production, this ETS(A,N,A) model assumes that production levels fluctuate in a predictable seasonal pattern without a long-term trend. This aligns well with agricultural production cycles that tend to repeat annually.

Forecast:

```
> hw.ZZZ.pred$mean
          Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct
2014 722.4634 710.3227 672.9659 645.1323 638.7748 645.1256 641.4450 692.1167 736.7834 813.1007
2015 722.4634 710.3227 672.9659 645.1323 638.7748 645.1256 641.4450 692.1167 736.7834 813.1007
2016 722.4634 710.3227 672.9659 645.1323 638.7748 645.1256 641.4450 692.1167 736.7834 813.1007
2017 722.4634 710.3227 672.9659 645.1323 638.7748 645.1256 641.4450 692.1167 736.7834 813.1007
          Nov      Dec
2014 820.2078 817.6791
2015 820.2078 817.6791
2016 820.2078 817.6791
2017 820.2078 817.6791
>
```

Question 4b:

```
> HW.ZZZ <- ets(prod.ts, model = "ZZZ")
> HW.ZZZ
ETS(M,N,A)

Call:
ets(y = prod.ts, model = "ZZZ")

  Smoothing parameters:
    alpha = 0.7228
    gamma = 1e-04

  Initial states:
    l = 773.363
    s = 103.0157 107.2804 99.5619 23.3131 -21.1053 -71.0627
           -69.9088 -75.1749 -66.2109 -34.9469 -3.1044 8.3428

  sigma:  0.0332

     AIC     AICc      BIC
3015.036 3017.070 3067.977
>
```

```
> #Model identified in 4A
> HW.ANA <- ets(prod.ts, model = "ANA")
> HW.ANA
ETS(A,N,A)

Call:
ets(y = prod.ts, model = "ANA")

  Smoothing parameters:
    alpha = 0.7105
    gamma = 1e-04

  Initial states:
    l = 771.4624
    s = 102.633 107.8419 98.9223 22.8155 -18.4778 -70.0518
           -67.3435 -75.2089 -67.5372 -38.0394 -3.1756 7.6213

  sigma:  24.4802

     AIC     AICc      BIC
3020.736 3022.770 3073.678
>
```

Holt-Winter's Forecast Using Full Dataset

To forecast monthly food production for 2018, two Holt-Winter's models were developed using the full dataset (1997–2017):

1. **Automated Model – ETS(M,N,A):**
   This model was identified by automated selection and includes a multiplicative error, no trend**,** and additive seasonality. It performed well in capturing stable seasonal patterns with a low sigma (0.0332).
2. **Specified Model – ETS(A,N,A):**
   Based on the model selected during training (Question 4a), the ETS(A,N,A) model was fit to the full dataset. It showed slightly higher information criteria (AIC = 3020.736) and a higher residual error (sigma = 24.4802).

Forecast:

```
> HW.ZZZ.pred <- forecast(HW.ZZZ, h = 12 , level = 0)
> HW.ZZZ.pred$mean
          Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct
2018 679.8990 668.4509 636.6052 605.3473 596.3829 601.6545 600.4930 650.4548 694.8668 771.1175
          Nov      Dec
2018 778.8395 774.5715
> HW.ANA.pred <- forecast(HW.ANA, h = 12 , level = 0)
> HW.ANA.pred$mean
          Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct
2018 679.3660 668.5659 633.7059 604.2061 596.5327 604.4005 601.6925 653.2665 694.5622 770.6671
          Nov      Dec
2018 779.5861 774.3774
>
```

Question 4c

```
> round(accuracy((snaive(prod.ts))$fitted, prod.ts), 3)
            ME   RMSE    MAE    MPE  MAPE  ACF1 Theil's U
Test set -3.93 48.394 38.897 -0.786 5.519 0.763     1.056
> round(accuracy(HW.ZZZ.pred$fitted, prod.ts), 3)
             ME   RMSE    MAE    MPE  MAPE  ACF1 Theil's U
Test set -0.559 23.836 17.572 -0.142 2.396 0.018     0.499
> round(accuracy(HW.ANA.pred$fitted, prod.ts), 3)
             ME  RMSE    MAE   MPE  MAPE  ACF1 Theil's U
Test set -0.557 23.79 17.652 -0.15 2.412 0.029     0.501
>
```

• **RMSE:** Between the three models, the HW model with automated selection (ETS(M,N,A)) has the lowest RMSE (23.836), followed closely by the specified ETS(A,N,A) model (23.79**),** while the seasonal naïve forecast performs the worst **(**48.394**).**

This means the Holt-Winter's models produce significantly more accurate monthly forecasts for U.S. food production than the seasonal naïve approach, with average deviations of around 23.8 million tons compared to 48.4 million tons.

• **MAPE:**The HW model (ETS(M,N,A)) again performs best with the lowest MAPE (2.396%), followed closely by ETS(A,N,A) (2.412%**),** both improving significantly over the seasonal naïve model (5.519%**).**

This shows that the Holt-Winter's models provide predictions that are, on average, within 2.4% of the actual production values.

Conclusion:
Based on both RMSE and MAPE, the Holt-Winter's model with automated selection

(ETS(M,N,A)) delivers the most accurate forecasts and outperforms the seasonal naïve and specified HW models. Therefore, it is the best exponential smoothing model for forecasting food production in this case study.

Final Model Comparison and Conclusion

When comparing the best models from both sections, we can clearly see a consistent improvement as we applied more advanced forecasting techniques.

In Question 3e, the two-level model (regression with trend and seasonality + trailing MA for residuals) performed significantly better than the basic regression and seasonal naïve models, achieving the lowest RMSE of 19.46 and MAPE of 2.05%**.** That was already a strong result and showed that layering techniques like correcting residuals helped us reduce error and make more reliable predictions.

Then in Question 4c, we moved into exponential smoothing models. While the Holt-Winter's model with automated selection (ETS(M,N,A)) was the best among the exponential smoothing group, its RMSE was slightly higher at 23.84 and MAPE was 2.40% still very good, but not as accurate as the two-level model from earlier.

So, It tells us that **each modeling step we applied was a genuine improvement**. The more we added thoughtful structure to the models like accounting for seasonality, then correcting residuals, and then using exponential smoothing the better the forecasts became. This steady progress is a really good sign that everything we've built so far is working correctly and logically. The models are improving because the steps make sense, not by luck.

Final Choice:
Even though both methods performed well, the two-level model from Question 3e is still the most accurate overall. So for predicting food production in 2018, I would go with the two-level model as the final forecasting method.